UNIT 5 · STANDARD 6.G.1

Area of Composite Figures

Lesson 5-5

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

An L-shaped room = a 12×8 rectangle joined to a 6×5 rectangle; total area = 96 + 30 = 126 sq ft

Composite Figure

A shape made by putting two or more simple shapes together.

Draw a dashed line across the L-shape to split it into two rectangles — now you can find each area separately

Decompose

To break a shape into smaller, simpler shapes.

T-shaped hallway: top rectangle = 30 sq ft, bottom rectangle = 28 sq ft → total = 30 + 28 = 58 sq ft

Add

Add up the areas of the smaller shapes to get the total.

A 14×10 pool with a 6×4 cutout: 140 − 24 = 116 sq ft of water surface

Subtract

Take away the area of a missing piece from a bigger shape.

Rectangle: $A = l \times w$; Triangle: $A = ½ \times b \times h$; use the right formula for each piece of a composite figure

Formula

A math rule written with symbols.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can find the area of a composite figure by adding or subtracting the areas of basic shapes.

1 Notice

Your architecture firm is calculating the floor area of an L-shaped room for a client's new home.

2 Key idea

I can find the area of a composite figure by adding or subtracting the areas of basic shapes.

3 Words I'll use

Composite Figure **Decompose** **Add** **Subtract** **Formula**

🤔 Think About It

- What does the shape of the room look like?
- Can you see simpler shapes inside the L-shape?
- What measurements are given for each part?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: An L-shaped room is made of two rectangles: 10 ft × 6 ft and 4 ft × 3 ft. What is the total area?

A. 72 sq ft
B. 60 sq ft
C. 12 sq ft
D. 72 ft

Step 1 Area 1 = 10 × 6 = 60 sq ft.

Step 2 Area 2 = 4 × 3 = 12 sq ft.

Step 3 Total = 60 + 12 = 72 sq ft.

✅ **Answer:** A. 72 sq ft

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: A rectangular patio is 15 ft × 10 ft with a 5 ft × 4 ft rectangular flower bed cut out. What is the remaining area?

A. 130 sq ft

B. 150 sq ft

C. 20 sq ft

D. 170 sq ft

Step 1 ______________________________

Step 2 ______________________________

Step 3 ______________________________

Answer: ______________________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Find the area of an L-shape made of a 5×3 rectangle and a 2×4 rectangle.

A. 23 sq units

B. 15 sq units

C. 8 sq units

D. 20 sq units

Show your work:

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The L-shaped room splits into a 12 ft by 8 ft rectangle and a 6 ft by 5 ft rectangle. Why is it easier to find the area after you decompose the L into two rectangles?

Level 1 support

Start here: We decompose the L-shape because we already know how to find a rectangle's area.

💡 **Need a hint?**

Sentence starters (optional)

We decompose the L-shape into ____ because ____.

Descomponemos la forma de L en ____ porque ____.

After I split it, I can find each area using ____.

Después de dividirla, puedo hallar cada área usando ____.

Word bank: composite figure | decompose | add | subtract | formula

Listen for: *Students explain that the L-shape is hard to measure directly, so breaking it into two rectangles lets them use A = l x w on each piece and add the results.*

Level 2

The wonder prompt asks if there is more than one way to split the L. Show a different decomposition and justify why both give the same total area.

- Another way to split it is ____, which gives ____ + ____.
- Both ways give the same total because ____.

EXPLORE

For the U-shaped pool you subtracted a 6 ft by 4 ft cutout from a 14 ft by 10 ft rectangle. How do you decide when to ADD areas and when to SUBTRACT?

Level 1 support

Start here: We add when shapes are joined and subtract when a piece is removed.

💡 **Need a hint?**

Sentence starters (optional)

I broke the figure into ____.
Dividí la figura en ____.

I subtracted the areas because a piece was ____, so 140 minus 24 equals ____.
Resté las áreas porque una parte fue ____, entonces 140 menos 24 es ____.

I add areas when ____, but I subtract when ____.
Sumo áreas cuando ____, pero resto cuando ____.

Word bank: composite figure | decompose | add | subtract | cutout

Listen for: *A strong answer states you subtract when a piece is cut out (pool = 140 - 24 = 116 sq ft) and add when shapes are joined, with a clear reason for each.*

Level 2

Could you find the U-shaped pool's area by ADDING rectangles instead of subtracting a cutout? Justify whether both strategies reach the same answer.

- I could add by splitting the pool into ____, which gives ____.
- Both strategies match because ____.

CONNECT

The T-shaped conference room has a top of 14 ft by 4 ft and a stem of 6 ft by 10 ft, and carpet costs $5 per sq ft. Talk through finding the total cost.

Level 1 support

Start here: Find each rectangle's area, add them, then multiply by the price.

💡 **Need a hint?**

Sentence starters (optional)

The top area is ____ square feet and the stem area is ____ square feet.
El área de arriba es ____ pies cuadrados y el área del tronco es ____ pies cuadrados.

I added the areas to get ____ square feet.
Sumé las áreas para obtener ____ pies cuadrados.

The cost is ____ times $5, which equals $____.
El costo es ____ por $5, que es $____.

Word bank: composite figure | decompose | add | subtract | formula

Listen for: *Students compute top = 56 sq ft, stem = 60 sq ft, total = 116 sq ft, and cost = 116 x $5 = $580, explaining the add-then-multiply order.*

Level 2

A house-shaped wall is a rectangle topped with a triangle. Explain how composite-figure thinking handles a figure made of two DIFFERENT shape types, not just rectangles.

- I would find the rectangle area with ____ and the triangle area with ____.
- Then I ____ the areas because ____.

Write About the Math

The Writing Revolution

I can explain my steps using the words composite figure, decompose, add, and subtract.

1. Kernel Sentence

subject + verb

Model: Composite Figure is a shape made by putting two or more simple shapes together.
Figura compuesta es una figura formada al juntar dos o más figuras simples.

Write a kernel sentence about composite figure. Use a subject and a verb.
Escribe una oración base sobre figura compuesta. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Composite Figure matters in math
Figura compuesta importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Composite Figure matters in math because ____.** *Figura compuesta importa en matemáticas porque ____.*
but *pero*	**Composite Figure matters in math, but ____.** *Figura compuesta importa en matemáticas, pero ____.*
so *entonces*	**Composite Figure matters in math, so ____.** *Figura compuesta importa en matemáticas, entonces ____.*

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about composite figure.
Di un hecho verdadero sobre composite figure.

Composite figure ____.

____

Question
Pregunta

Ask a question about composite figure.
Haz una pregunta sobre composite figure.

How does ____ ?
¿Cómo ____ ?

____

Exclamation
Exclamación

Show excitement about composite figure.
Muestra entusiasmo sobre composite figure.

Wow, ____ !
¡Guau, ____ !

____

Command
Mandato

Tell a partner what to do with composite figure.
Dile a un compañero qué hacer con composite figure.

First, ____ .
Primero, ____ .

____

4. Explain Your Reasoning

use a sentence starter

We decompose the L-shape into ____ because ____.
Descomponemos la forma de L en ____ porque ____.

After I split it, I can find each area using ____.
Después de dividirla, puedo hallar cada área usando ____.

I broke the figure into ____.
Dividí la figura en ____.

____

____

____

Try It

Solve on your own. Check the answer key when you are done.

1. What is the best first step to find the area of a composite figure?

A. Break it into simple shapes like rectangles and triangles

B. Measure only one side

C. Add all the side lengths

D. Multiply the two longest sides

Show your work:

2. A garden is a 12×5 rectangle plus a triangle with base 12 and height 4 on top. What is the total area?

A. 84 sq units

B. 60 sq units

C. 24 sq units

D. 72 sq units

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A T-shaped room can be decomposed in two different ways. Show both ways and prove they give the same total area. Use a T-shape where the top is 14 ft × 4 ft and the stem is 6 ft × 10 ft.

Sentence starter: Way 1: I split it into ____ and ____. Areas: ____ + ____ = ____. Way 2: I used a ____ rectangle and subtracted ____. Areas: ____ − ____ = ____. Both give ____ sq ft.

Show your work:

____

____

____

____

Reflect — Exit Ticket

A composite figure is made of a 9 ft x 7 ft rectangle and a 3 ft x 4 ft rectangle joined together. What is the total area?

A. 75 sq ft

B. 63 sq ft

C. 12 sq ft

D. 75 ft

Your answer:

____

____

____

Answer Key & Teacher Guide

1. **We Try (notes):** A. 130 sq ft *— Patio = 15 × 10 = 150 sq ft. Cutout = 5 × 4 = 20 sq ft. Remaining = 150 − 20 = 130 sq ft.*
2. **You Try (notes):** A. 23 sq units *— 5 × 3 = 15. 2 × 4 = 8. Total = 15 + 8 = 23 sq units.*
3. **Try It 1:** A. Break it into simple shapes like rectangles and triangles *— Decompose the figure into simple shapes, find each area, then add or subtract.*
4. **Try It 2:** A. 84 sq units *— Rectangle 12 × 5 = 60; triangle ½ × 12 × 4 = 24. Total = 84 sq units.*
5. **Exit Ticket:** A. 75 sq ft *— Area 1 = 9 x 7 = 63 sq ft. Area 2 = 3 x 4 = 12 sq ft. Total = 63 + 12 = 75 square feet.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Composite Figure is a shape made by putting two or more simple shapes together.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).